

CO 2 AT1 SCENNARIO BASED QUESTIONS

Q1 Student Registration System

10 marks

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

- Enumerate three Collection interfaces suitable for this system.
- Design a Collection structure (diagram/representation) to store students and their enrolled courses.
- Write a Java program using HashSet and HashMap with Generics to:
 - Add students
 - Map students to courses
 - Display data using Iterator

Your Code

```
1 a.util.*;
2 == Student{
3 static void main(String[] args){
4 (a) three collection interface suitable for this system;
5 1. Set-stores unique students
6 2. List-stores multiple courses
7 3. Map-maps students to their courses
8
9 (b) collection structures:
10 HashSet<String> students
11 HashMap<String, ArrayList<String>> map
12 student registration system
13 |
14   HashSet
15 |
16   nisha, hanu, kannan
17
18   HashMap
19 student->courses
20 nisha->java, computer
21 hanu->python, physics
22 kannan->chemistry, biology
23
24 (c) implementation using HashSet, HashMap, Generics
25 HashSet<String> students = new HashSet<>();
26 students.add("nisha");
27 students.add("hanu");
28 students.add("kannan");
29 HashMap<String, ArrayList<String>> map = new HashMap<>();
30 p.put("nisha", new ArrayList<>(Arrays.asList("java", "computer")));
31 p.put("hanu", new ArrayList<>(Arrays.asList("python", "physics")));
32 p.put("kannan", new ArrayList<>(Arrays.asList("chemistry", "biology")));
33
34 Iterator<String> it = students.iterator();
35
36 while (it.hasNext()){
37   String student = it.next();
38   System.out.println("Student: " + student);
39   System.out.println("Courses: " + map.get(student));
40   System.out.println();
41
42 }
```

Input

stdin input

Output

System error.

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

- List three Collection classes applicable for cart management.
- Represent how cart items are stored using a List-based structure.
- Write a Java program using ArrayList and Iterator to:

- Add items
- Remove an item
- Display all items

Your Code

```
1 import java.util.*;
2 public class Cart{
3     public static void main(String[]args){
4         // (a) three collection
5         // 1. arraylist
6         // 2. linkedset
7         // 3. hashset
8
9         // (b) list based structure
10        // ArrayList<String> cart = new ArrayList<>();
11        //
12        // cart
13        // |
14        // +---laptop
15        // +---mouse
16        // +---keyboard
17        //
18        // items are stored in order and can be accessed
19
20        // (c) add, remove and display
21        ArrayList<String> cart = new ArrayList<>();
22        cart.add("laptop");
23        cart.add("mouse");
24        cart.add("keyboard");
25        cart.add("headphones");
26        cart.remove("mouse");
27        System.out.println("items in cart");
28        Iterator<String> it = cart.iterator();
29        while (it.hasNext()){
30            System.out.println(it.next());
31        }
32    }
33 }
```

 Run Save

Input

Output

System error.

Q3 Library Management System

10 marks

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

- (a) Name three Map implementations in Java.
- (b) Design a schema-like representation using Map for storing ISBN and book details.
- (c) Write a Java program using HashMap to:
 - Insert book details
 - Retrieve a book
 - Display all books

Your Code

```
1 import java.util.*;
2 public class library{
3     public static void main(String[] args){
4         // (a) three map implementation
5         // 1. hashmap
6         // 2. treemap
7         // 3. linkedhashmap
8
9         // (b) schema representation:
10        // hashmap<integer, string>
11
12        // ISBN --> book details
13        // 101 --> java
14        // 1-2 --> c
15        // 103 --> python
16
17        // (c) implementation of hashmap program
18        HashMap<Integer, String> books = new HashMap<>();
19        books.put(101, "java");
20        books.put(102, "c");
21        books.put(103, "python");
22        System.out.println("book with ISBN 102: " + books.get(102));
23        System.out.println("\nLibrary books:");
24        for (Map.Entry<Integer, String> entry : books.entrySet()) {
25            System.out.println("ISBN: " + entry.getKey() + "book: " + entry.getValue());
26        }
27    }
28 }
29 }
```

 Run Save**Input**

stdin input

Output

Q4 Employee Data Processing (10 Marks)

10 marks

Suppose that a company processes employee records and filters employees based on salary.

- (a) List three advantages of Generics in Collections.
- (b) Represent how employee data is stored using List with Generics.
- (c) Write a Java program using Iterator to:
 - Store employee objects
 - Display employees with salary > 50,000

Your Code

```
1 import java.util.*;
2 class Emp{
3     String name;
4     int salary;
5     Emp(String name,int salary){
6
7         this.name=name;
8         this.salary=salary;
9     }
10 }
11 }
12 public class 192521301RC{
13     public static void main(String[]args){
14         //(a)three advantages
15         //1. provides type safety
16         //2. detects errors at compile time
17         //3. avoids explicit type
18
19         //(b) employee data stored using list with generics:
20         //arraylist<emp>list=new arraylist<emp>();
21         //it stores emp objects in a type safe manner:
22
23         //(c)store employee objects and display salary>50000
24         ArrayList<Emp>list=new ArrayList<>();
25         list.add(new Emp("nisha",60000));
26         list.add(new Emp("hanu",45000));
27         list.add(new Emp("priya",70000));
28
29         Iterator<Emp>it=list.iterator();
30         while(it.hasNext()){
31             Emp e=it.next();
32             if(e.salary>50000){
33                 System.out.println("name:"+e.name+"salary:"+e.salary);
34             }
35         }
36     }
37 }
```

 Run Save**Input**

stdin input

Output

System error.

Q5 Online Voting System

10 marks

Suppose that an online voting system ensures one vote per user and counts votes efficiently.

- (a) Identify three suitable Collection types for this system.
- (b) Design a structure using Set and Map for voters and vote counting.
- (c) Write a Java program using HashSet and HashMap to:

- Record votes
- Prevent duplicate voting
- Display results

Your Code

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         // (a) three suitable collection types:
5         // 1. HashSet
6         // 2. HashMap
7         // 3. ArrayList
8
9         // (b) structure:
10        // HashSet<String> voters -> stores unique voter ids
11        // HashMap<String, Integer> votes -> stores candidate and vote count
12
13        // (c) online voting system
14        HashSet<String> voters = new HashSet<>();
15        HashMap<String, Integer> votes = new HashMap<>();
16        String[] voterID = {"V101", "V102", "V101"};
17        String[] candidate = {"nisha", "hanu", "kannan"};
18        for (int i = 0; i < voterID.length; i++) {
19            if (voters.contains(voterID[i])) {
20                System.out.println("duplicate vote are not allowed");
21            }
22            else {
23                voters.add(voterID[i]);
24                votes.put(candidate[i], votes.getOrDefault(candidate[i], 0));
25                System.out.println("vote recorded");
26            }
27        }
28        System.out.println("\n Voting results");
29        for (String c : votes.keySet()) {
30            System.out.println(c + " : " + votes.get(c));
31        }
32    }
33 }
34
35
36
37
```

Input

stdin input

Output

System error.